FairEdge

D4F Assignment

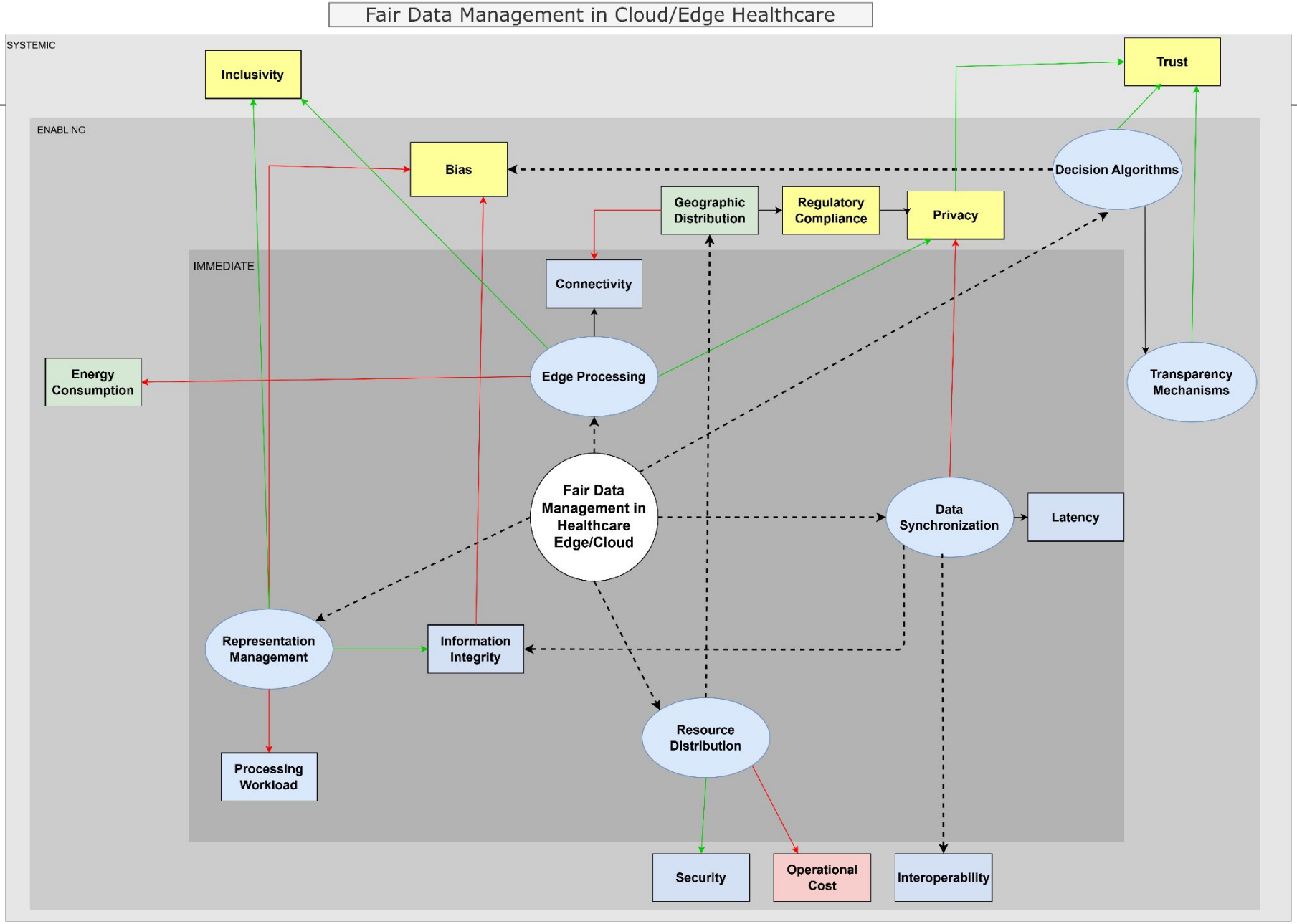
Use case: FairEdge

Maria is a doctor at a hospital that works with a cloud-based patient monitoring system. This system uses cloud computing to collect health data from patients through wearables and processes this data in the cloud.

There are a few issues with reliability in the collection and analysis of data. Patients from rural areas have more trouble connecting to the system at times which results in their data being less accurate compared to patients from less rural areas. Local privacy laws also mean some patients' data cannot be processed or stored centrally.

The FairEdge system will address these problems by combining edge computing with the current cloud computing solution. Local data processing will be prioritized on edge devices in rural areas. The system will adapt data processing based on reliability of the data. It will also include transparent documentation about how the system makes decisions.

Definition of Fairness: Decision Map



Definition of Fairness: Decision Map (cont.)

System Features

Edge Processing: The key feature that makes rural patients' data more reliable, though it increases energy consumption.
Representation Management: This is about keeping all data reliable for all patients, regardless of location.
Data Synchronization: Necessary to keep edge and cloud computing working hand in hand, but creates latency issues.
Decision Algorithms: These are there for diagnostics, could create potential bias.
Transparency Mechanisms: These mechanisms are there to build trust and to gain consent.
Resource Distribution: The main system that manages resources across different areas to keep fairness high.

Concerns

Bias and Inclusivity: The data from rural patients is currently less reliable which creates fairness issues.
Geographic Distribution: It is a concern that the location of patients affects data quality and access to the system.
Connectivity: This is worse in rural areas, so we need special care to keep everything fair.
Information Integrity: All information must be accurate, especially given the context of the data.
Privacy and Regulatory Compliance: Our system has to comply with different laws and regulations across areas.
Security: The patient data must always be protected regardless of the location.
Interoperability: All facilities must be able to participate in the system.
Processing Workload: Fairness must be maintained, so the processing workload is higher than otherwise. This is a performance concern.
Operational Costs: This is a cost concern related to the implementation of our fairness system.
Latency: A technical issue affecting access to real-time insights.
Energy Consumption: This is about the environmental impact of the new system architecture.
Trust: Both patients and healthcare providers must have trust in order to implement this system.

Definition of Fairness: SQ model

FILL IN YOUR OWN QA NAME	FILL IN YOUR OWN QA DEFINITION	FILL IN THE EXTERNAL SOURCE OF THE DEFINITION (if applicable)	SUSTAINABILITY DIMENSION OF THE QA				FILL IN MEASURES IN USE (if applicable)
QUALITY ATTRIBUTE (QA)	DEFINITION	REFERENCE	TECHNICAL	ENVIRONMENTAL	ECONOMIC	SOCIAL	DEFINITION of METRIC/KPI
Representation Management	The degree to which all patient data is equally represented in the system regardless of origin, connectivity, or demographic factors.					X	Goal: Maximize data representation equality CSF: Balance data influence from all regions Metric: (Data completeness % in worst connected region) / (Data completeness % in best connected region) × 100
Bias	The degree to which the system prevents favoritism or discrimination against patient groups based on their location					X	
Trust	The degree to which healthcare providers and patients have confidence in the system's recommendations and processes.					X	
Information Integrity	The property of data to remain accurate, complete and representative throughout its lifecycle, regardless of geographic origin of patient data.		X				
Inclusivity	The ability of the system to incorporate and address needs of all patient populations regardless of location, connectivity, or other limiting factors.					X	
Privacy	The degree to which the collection, storage, and use of patient health data is protected from unauthorized access and respects patient autonomy.					X	
Connectivity	The robustness of network connections between system components and the ability to function with intermittent or limited connectivity.		X				
Resource Distribution	The allocation of computational and storage resources across geographic locations to ensure equitable system performance.		X				
Geographic Distribution	The spatial arrangement of system components and data storage across different physical locations.			X			
Operational Cost	The financial resources required to maintain equitable operation of the distributed healthcare system.				X		
Processing Workload	The degree to which the system manages computational burden across the distributed healthcare environment		X				
Latency	The degree to which the system minimizes time delays during data transfer and processing between components		X				
Security	The degree to which the system protects data and operations from unauthorized access or malicious activities		X				
Energy consumption	The environmental impact of distributed processing architecture.			X			

Definition of Fairness: SQ model (cont.)

Our SQ model categorizes 15 quality attributes for healthcare edge/cloud fairness across four sustainability dimensions:

Social Dimension
Representation Management, Bias, Trust, Inclusivity, and **Privacy** form the social aspect of our definition of fairness. These attributes aim to give all patients equal care regardless of location. There is an emphasis here on avoiding systemic disadvantages for more rural patients.

Technical Dimension
Information Integrity, Connectivity, Latency, **Resource distribution, Processing workload,** and **Security** are about the technical issues with data quality and system access. Since there are different environments and locations at play, it is crucial to keep rural patients' data accurate and accessible.

Economic Dimension
Operational Cost is the only economic concern. The goal here is to minimize economic impact of the distribution of resources.

Environmental Dimension
Geographic Distribution and **Energy Consumption** minimize the environmental impact and maintain the fairness standards.

These quality attributes are our approach to fairness with the FairEdge system. They address the issues that come up in our use case where patients in rural areas experience worse data quality due to connectivity issues. By mapping these attributes across the different sustainability dimensions, we can make our fairness definition comprehensive and still address all relevant aspects of Maria's healthcare monitoring system.

Fairness perspective

	Description
DESIRED QUALITY	***Representation Management*** *The degree to which all patient data is equally represented in the system regardless of origin, connectivity, or demographic factors.*
CONCERNS	***Representation Management*** *involves working towards fair representation of all patient groups regardless of geolocation* ***Bias Prevention*** *involves preventing systemic disadvantages for rural patients in medical processes* ***Connectivity Fairness*** *involves facilitating various connection types without penalizing the rural areas* ***Information Integrity*** *involves working towards identical data accuracy across all geolocations* ***Geographic Distribution*** *involves mitigating location disparities in system access and data quality* ***Privacy & Regulatory Compliance*** *requires adherence to data laws and regulations of different areas with guaranteed fairness* ***Resource Distribution*** *involves providing equitable distribution of computing resources to all geolocations*
ARCHITECTURAL TACTIC(S)	***Adaptive Edge Processing Prioritization*** *This tactic can be used to assign processing resources to edge devices dynamically based on connectivity metrics, prioritizing rural areas with low connectivity to deliver an equal representation and quality of information to every patient group*
POSSIBLE KPI(S)	***Geographic Data Representation Gap (GDRG):*** *Measures the difference in data completeness percentage between the best and worst-connected patient regions. Target: less than 5%*
PROBLEMS AND PITFALLS	***Energy Consumption*** *may be higher in total because of edge processing in rural areas* ***Operational Costs*** *could be greater with more edge resources in rural areas* ***Synchronization Challenges*** *may occur when synchronizing data between distributed nodes with varying connectivity*

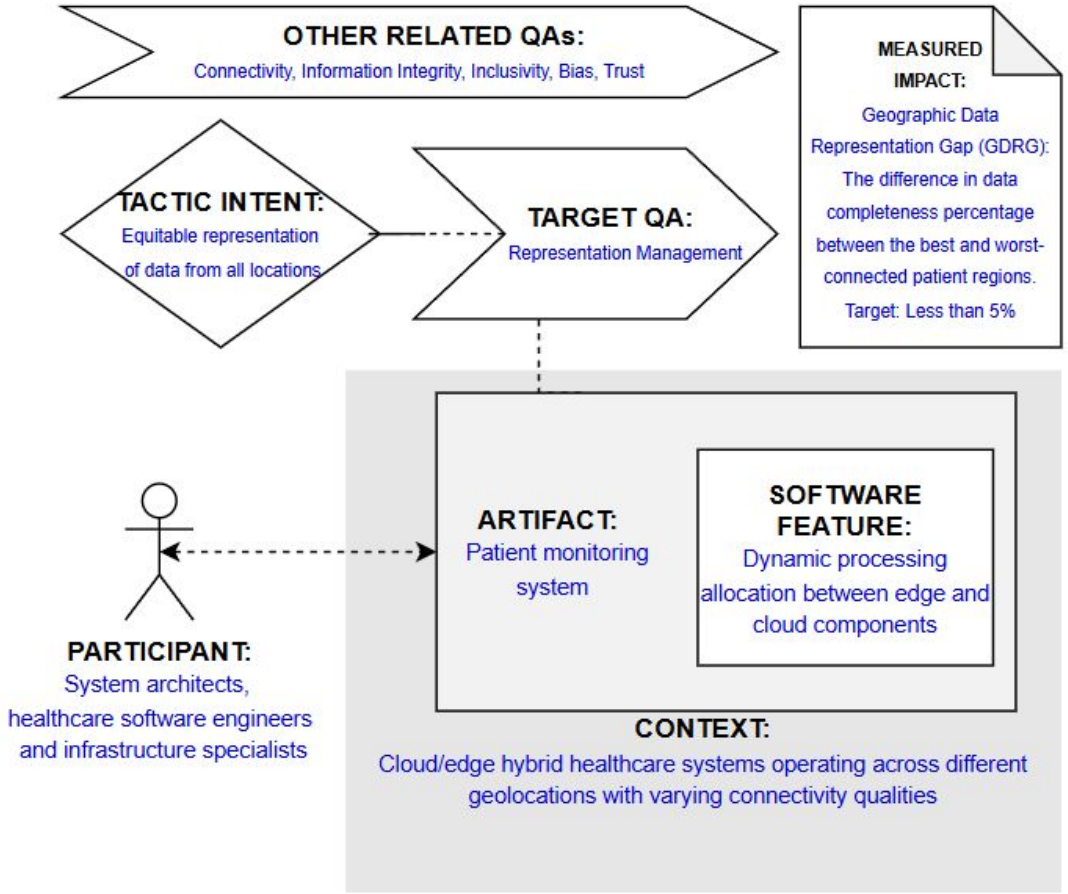
Definition of KPI*

Representation Management					
Goal	**Critical Success Factor (CSF)**	**Key Performance Indicator (KPI)**		**Metric(s)**	**Measure(s)**
Maximize data representation equality across geographic regions	***Balance data influence from all regions regardless of connectivity quality***	***Geographic Data Representation Gap (GDRG)***		***(Data completeness % in worst connected region) ÷ (Data completeness % in best connected region) × 100***	***Percentage ratio (0-100%)***
		Target	**Action(s)**		
		< 5% difference between best and worst-connected patient regions	• ***Implement adaptive edge processing prioritization in rural areas*** • ***Establish data quality monitoring across all geographic regions*** • ***Put more edge computing tools in places where connectivity is weak.***		

Architectural Tactic(s) for Fairness*

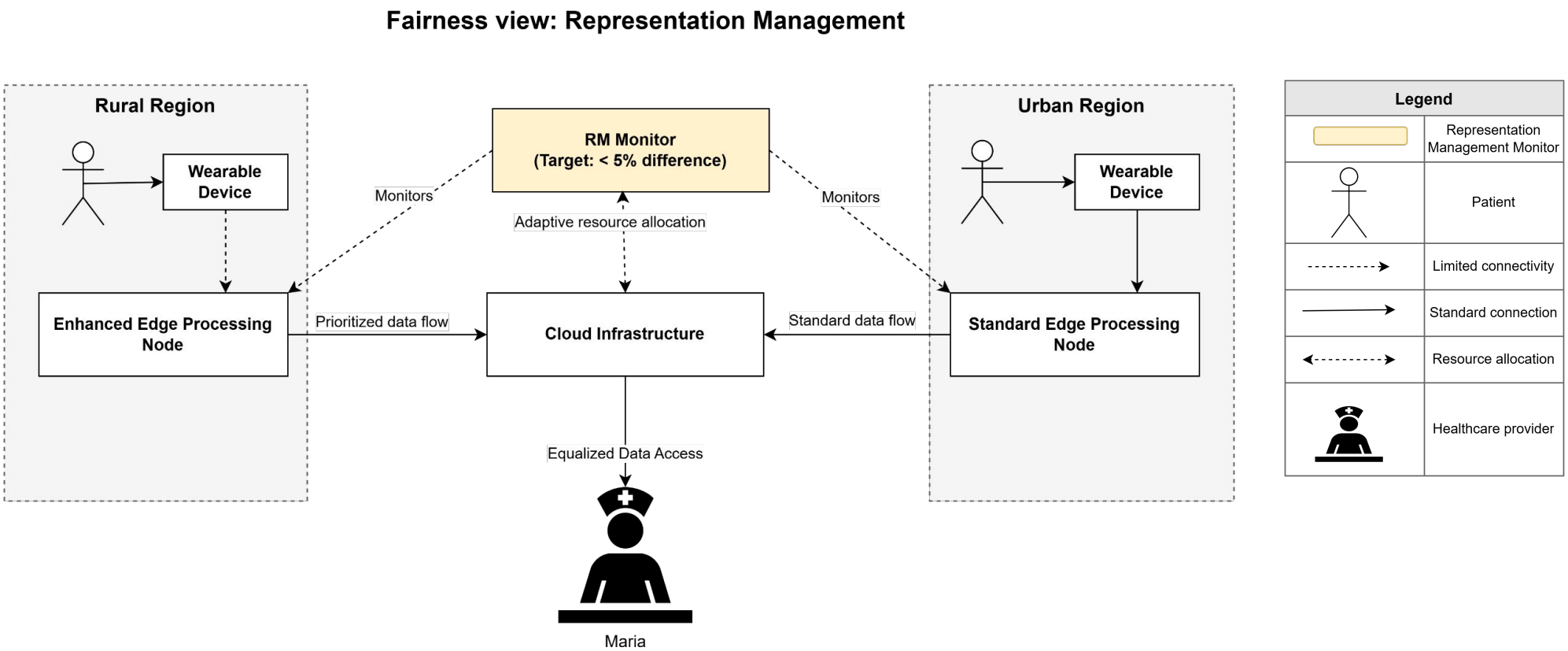
title: Adaptive Edge Processing Prioritization
tags: edge-computing, measured
t-sort: Awesome Tactic
t-type: Architectural Tactic
categories: resource-adaptation
t-description: Dynamically allocate processing power to edge devices based on geographic connectivity metrics with a bias towards rural areas with low connectivity in order to provide equal representation and quality of data for all groups of patients
t-participant: System architects, healthcare software engineers, and infrastructure specialists
t-artifact: Patient monitoring system
t-context: Cloud/edge hybrid healthcare systems operating across different geolocations with varying connectivity qualities
t-feature: Dynamic processing allocation between edge and cloud components
t-intent: In order to get an equitable representation of patient data from all geolocations, rural areas with connectivity challenges must be focused on. The goal here is to reduce bias in medical data and decisions
t-targetQA: Representation Management
t-relatedQA: Connectivity, Information Integrity, Inclusivity, Bias, Trust
t-measuredimpact: Geographic Data Representation Gap (GDRG): The difference in data completeness percentage between the best and worst-connected patient regions.
Target: GDRG < 5%
t-source: Ramanathan, S., Ramasundaram, M., Mohd, T. K., Gunasekaran, H., & Pineda-Briseño, A. (2023). *Edge Computing with IoT devices in Healthcare: Concepts, Tools, Techniques, and Use cases. In Edge Computing in Healthcare.*

Tactics, A. a. D. (2022). *Apply edge computing. Awesome and Dark Tactics.* https://s2group.cs.vu.nl/AwesomeAndDarkTactics/2022-04-20-apply-edge-computing
t-source-doi: 10.1201/9781003363361-1



This tactic addresses our fairness definition by creating a technical solution to the main problem. This tactic helps work towards equal data representation in the system regardless of where patients live or their connection quality.

Fairness view: Representation Management view



Fairness view (cont.)

The diagram in previous slide shows our design for the FairEdge system that uses Adaptive edge processing prioritization to make data more fair between rural and urban areas. The main problem in our use case is that patients in rural areas have worse data quality than patients in urban areas because of connectivity issues. We made this view to show how our system fixes that problem.

In our architecture, we handle fairness through these mechanisms:

- **Different types of edge processing:** Rural areas get Enhanced Edge Processing Nodes with more resources to process data locally when connectivity is poor. These nodes help maintain good data even when internet connection is bad.
- **Priority for rural data**: Data coming from rural areas gets priority when going to the cloud. This helps because if connections are intermittent, the system makes sure rural data doesn't get left behind.
- **Representation Management Monitor:** Our monitoring system tracks the difference between data quality in different regions. When the gap gets close to 5%, the system automatically gives more resources to the areas that need it most.

This architecture connects to our fairness perspective by tackling multiple dimensions:

- **Social dimension**: Makes sure all patients have equal representation
- **Technical dimension:** Fixes connectivity issues for rural areas
- **Economic dimension**: Uses resources efficiently without too much cost
- **Environmental dimension:** Distributes processing in a balanced manner

For Maria (the doctor from our use case), this means she gets good quality data from all her patients no matter where they live. When she looks at patient data, she won't see big differences in quality between urban and rural patients. This means she can make fair decisions based on equal information

Our design transforms the unequal data collection problem into a fair system by using adaptive resource allocation. This keeps the data quality difference below our 5% target, which means better and more fair healthcare for everyone.